Analytical Velocity / Yaw-Rate Limits, Envelope Geometry, and Cross-Axis Coupling of a Mecanum Platform under Steady-State LuGre Friction

Mecanum–PINN dynamics notes
youBot / slip–spin–LuGre simulator (Julia), `:lugre adamov coupling`

June 5, 2026 (rev. acceleration envelope, June 6, 2026)

Abstract

These notes collect the closed-form results derived for the four-wheel Mecanum platform of the slip–spin–LuGre simulator. We give (i) the 3-DOF equivalent (augmented) plant model the controller and observer act on; (ii) the analytical caps on the longitudinal, lateral, and yaw-rate axes and the physically distinct mechanisms that set them; (iii) the structural shape of the no-slip / trackable velocity envelope in the translational plane ($\dot{\psi} = 0$); (iv) the extension of the caps to a conservative *velocity–acceleration* envelope built per wheel on the same friction circle (§3.4); and (v) the origin and quantification of the cross-axis coupling. All quantities use the case-1 parameter set. A Word-compatible equation appendix is provided for transcription into the manuscript.

1 System and notation

The platform carries four Mecanum wheels in the O-configuration with roller-axis angles

$$\delta = \left(-\tfrac{\pi}{4}, \tfrac{\pi}{4}, \tfrac{\pi}{4}, -\tfrac{\pi}{4}\right), \quad \cos\delta_i = \tfrac{1}{\sqrt{2}}\ \forall i, \quad \cos^2\delta_i = \tfrac{1}{2}. \tag{1}$$

Wheel centres are at $p_x = (h, h, -h, -h)$, $p_y = (l, -l, l, -l)$. The body velocity is $\vec{v} = (V_x, V_y, \dot{\psi})$; the centre of mass is offset from the geometric centre by (a_X, a_Y). The parameters used throughout are listed in Table 1.

Sym	Value	Meaning	Sym	Value	Meaning
h	0.235 m	half-length	μ	0.5	Coulomb coeff.
l	0.15 m	half-width	$s = \mu_s/\mu_c$	1.1	stiction ratio
R	0.05 m	wheel radius	v_s	0.01 m/s	Stribeck vel.
R_a	0.0355 m	roller axle dist.	p_1	0.11	wheel viscous
$R - R_a$	0.0145 m	—	p_2	5.78×10^{-3}	roller viscous
m	30 kg	body mass	$\tau_{\max}$	10 N m	torque cap
m_s	35.6 kg	total mass	a_X	0.016 m	COM offset (long.)
$N_{tot} = m_s g$	349.2 N	total normal	a_Y	−0.026 m	COM offset (lat.)
J_w	5.87×10^{-3}	wheel inertia	χ	$\{0, .., .008\}$	contact-spot

Table 1: Case-1 parameter set (from `PlatformParams / LuGreParams`).

Contact slip kinematics. With the integrated roller-spin rate $\dot{\gamma}_i$ and the wheel rate ω_i, the per-contact slip velocities are, at the representative roller phase $\tilde{\varphi}_i \to 0$,

$$V_{px,i} = V_x - \dot{\psi}\, p_{y,i} - \omega_i R + \dot{\gamma}_i \sin\delta_i (R_a - R), \tag{2}$$
$$V_{py,i} = V_y + \dot{\psi}\, p_{x,i} + \dot{\gamma}_i \cos\delta_i (R - R_a). \tag{3}$$

The decisive structural fact is that the wheel nulls the body-x slip (its lever is $p_y = \pm l$) and the passive roller-spin state $\dot{\gamma}_i$ nulls the body-y slip (its lever is $p_x = \pm h$); the roller inertia $J_{\text{roller}} = 10^{-6}$ is driven solely by contact friction against the viscous drag p_2.

Wheel and roller dynamics (slip retained, for reference). Away from the no-slip reduction of §2, the per-wheel and per-roller equations of Adamov & Saipulaev (their Eq. 3, lines 4–5) are

$$J_1 \ddot{\varphi}_i = M_i - p_1 \dot{\varphi}_i - F_{x,i} R, \tag{4}$$
$$J_r \ddot{\gamma}_i = -p_2 \dot{\gamma}_i - F_{x,i} \sin\delta_i (R - R_a \cos\tilde{\varphi}_i) - F_{y,i} \cos\delta_i (R_a - R\cos\tilde{\varphi}_i) + \cdots, \tag{5}$$

with the platform rows (x, y, ψ) driven by $\sum_i F_{x,i}$, $\sum_i F_{y,i}$, and $\sum_i (F_{y,i}\rho_{Xi} - F_{x,i}\rho_{Yi} + M_{z,i})$. The motor enters only through the wheel equation; the roller is *passive* — $\dot{\gamma}_i$ is sustained only by contact terms against p_2. This actuated-wheel / passive-roller asymmetry splits the caps in §3.

Steady-state friction. The LuGre/Stribeck steady force per contact is a bounded vector aligned with the slip,

$$\vec{F}_i = -N_i\, g_t(\|V_p\|)\, \frac{V_{p,i}}{\|V_{p,i}\|}, \quad g_t = \mu_c \left[1 + (s-1) e^{-(\|V_p\|/v_s)^2}\right], \tag{6}$$

so $\|\vec{F}_i\|$ ranges from $N_i \mu_s$ (static) down to $N_i \mu_c$ (kinetic): each contact has an essentially isotropic friction circle of radius $\sim N_i \mu$.

Roller-frame force decomposition. Every contact force lives in the plane of its roller, so the natural basis is the roller frame — the axle direction $\hat{e}_i = (\cos\delta_i, \sin\delta_i)$ and the rolling direction $\hat{n}_i = (-\sin\delta_i, \cos\delta_i)$ — in which

$$\vec{F}_i = F_{\parallel,i}\, \hat{e}_i + F_{\perp,i}\, \hat{n}_i, \qquad F_{\parallel,i}^2 + F_{\perp,i}^2 \le (\mu N_i)^2. \tag{7}$$

The two components have distinct owners. Along the axle the roller cannot roll, so $F_\parallel$ is the wheel traction set by the motor (carried with the bearing p_1). Along the rolling direction the roller spins freely against p_2, so $F_\perp = p_2 \dot{\gamma}_i / (R - R_a)$ is the roller-sustaining drag, supplied by contact friction because the roller is passive. This is the p_1/p_2 split made geometric: $F_\parallel \leftrightarrow$ motor, $F_\perp \leftrightarrow$ contact. A longitudinal command leaves $F_\perp = 0$ and the circle empty (a wheel/motor limit); a lateral or yaw command forces $F_\perp \ne 0$ and fills it (a roller/contact limit).

2 Equivalent (augmented) plant model

The controller and disturbance observer are built on a 3-DOF rigid-body equivalent in which the wheel/roller rotational inertias are reflected into the body and the lumped contact dissipation is written as equivalent viscous and Coulomb terms. With $\vec{v} = (V_x, V_y, \dot{\psi})$,

$$M_{aug}\, \dot{\vec{v}} + C(\dot{\psi})\, \vec{v} + D_{eq}\, \vec{v} + \vec{f}_c(\vec{v}) = \vec{F}_p. \tag{8}$$

The augmented mass matrix reflects J_w through the wheel kinematics and keeps the COM coupling,

$$M_{aug} = \begin{bmatrix} \tilde{m} & 0 & -ma_Y \\ 0 & \tilde{m} & ma_X \\ -ma_Y & ma_X & I_\psi \end{bmatrix}, \quad \tilde{m} = m_s + \frac{4J_w}{R^2}, \quad I_\psi = I_s + \frac{4(l+h)^2}{R^2} J_w. \tag{9}$$

The gyroscopic/centrifugal vector (the only translation–yaw coupling once M_{aug} is formed) is

$$C(\dot{\psi})\,\vec{v} = \begin{bmatrix} -m_s\dot{\psi}V_y - ma_X\dot{\psi}^2 \\ m_s\dot{\psi}V_x - ma_Y\dot{\psi}^2 \\ m\dot{\psi}(a_X V_x + a_Y V_y) \end{bmatrix}. \tag{10}$$

Equivalent damping from the no-slip (Model N) reduction. In Adamov & Saipulaev the dissipation lives only in the per-wheel/roller equations — base–wheel joint $p_1\dot{\varphi}_i$ and wheel–roller joint $p_2\dot{\gamma}_i$ — and the platform equations carry no explicit body damping. Imposing the no-slip constraints $V_{PiX} = V_{PiY} = 0$ at $\tilde{\varphi}_i = 0$ and solving,

$$\dot{\gamma}_i = -\frac{V_y + \rho_{Xi}\dot{\psi}}{\cos\delta_i (R - R_a)}, \qquad \dot{\varphi}_i = \frac{1}{R}\Big[V_x - \rho_{Yi}\dot{\psi} + (V_y + \rho_{Xi}\dot{\psi})\tan\delta_i\Big]. \tag{11}$$

With the Rayleigh dissipation $\mathcal{F} = \frac{1}{2}\sum_i (p_1\dot{\varphi}_i^2 + p_2\dot{\gamma}_i^2)$ the equivalent body damping is the Hessian $D_{eq} = \sum_i (p_1 \nabla\dot{\varphi}_i \nabla\dot{\varphi}_i^\top + p_2 \nabla\dot{\gamma}_i \nabla\dot{\gamma}_i^\top)$, which (all off-diagonals cancelling) is

$$D_{eq} = \mathrm{diag}\left(\underbrace{\tfrac{4p_1}{R^2}}_{d_x},\ \underbrace{\tfrac{4p_1}{R^2} + \tfrac{8p_2}{(R-R_a)^2}}_{d_y},\ \underbrace{\tfrac{4p_1(l+h)^2}{R^2} + \tfrac{8p_2 h^2}{(R-R_a)^2}}_{d_\psi} \right) \approx \mathrm{diag}(176, 396, 38). \tag{12}$$

The p_1/p_2 split that drives every cap. Each entry separates into a wheel-joint (p_1) part paid by the motors (at steady state $M_i = F_{Xi}R + p_1\dot{\varphi}_i$, so bearing drag is covered by motor torque and never loads the circle) and a roller-joint (p_2) part that must be supplied by contact friction because the roller is passive. Consequently every friction-limited cap (§3) is set by the p_2 part of its axis, while the p_1 part sets the actuator/power demand: V_x has no p_2 term (motor-limited, large), whereas V_y and $\dot{\psi}$ are governed by their p_2 parts (friction-limited, small). The control wrench is the wheel-torque map

$$\vec{F}_p = \big(\tfrac{1}{R}\textstyle\sum_i \tau_i,\ \tfrac{1}{R}(-\tau_1 + \tau_2 + \tau_3 - \tau_4),\ \tfrac{l+h}{R}(-\tau_1 + \tau_2 - \tau_3 + \tau_4)\big). \tag{13}$$

3 Velocity and yaw-rate caps

3.1 Longitudinal cap — torque/drag limited ($\dot{\psi} = 0$)

At constant V_x the rollers need not spin ($\dot{\gamma}_i = 0$, hence $F_{\perp,i} = 0$), and a wheel rolling at steady speed needs no traction ($F_{\parallel,i} \to 0$): the circle is empty. The only steady resistance is the bearing torque $p_1\omega_i$, which saturates at $\tau_{\max}$:

$$V_{x,cap} = \frac{\tau_{\max} R}{p_1} = \frac{10 \times 0.05}{0.11} \approx 4.55 \text{ m/s}. \tag{14}$$

This is a *soft* cap: the circle never binds. The transient (maneuver) limit is the aggregate friction circle,

$$a_{x,\max} = \frac{(\sum_i N_i)\mu}{m_s} = \frac{349.2 \times 0.5}{35.6} \approx 4.9 \text{ m/s}^2. \tag{15}$$

3.2 Lateral cap — friction-circle limited ($\dot\psi = 0$)

For pure strafe the rollers must spin to null $V_{py,i}$, at $\dot\gamma_i = -V_y/(\cos\delta_i(R-R_a))$, so each contact carries the roller-sustaining drag

$$F_{\perp,i} = \frac{p_2\dot\gamma_i}{R-R_a} = \frac{\sqrt{2}\,p_2}{(R-R_a)^2}V_y \quad (\text{along } \hat n_i). \tag{16}$$

This $F_\perp$ is the same on every wheel and sums to a pure $-y$ drag $2\sqrt{2}\,F_\perp$. The steady balance forces the tractions to supply this in y and nothing net in x, which — by the symmetric $\pm$ pattern — pins $|F_{\parallel,i}| = F_{\perp,i}$ at every wheel. The two roller-frame components being equal, each wheel's resultant has magnitude

$$\|\vec F_i\| = \sqrt{2}\,F_{\perp,i} = \frac{2p_2}{(R-R_a)^2}V_y \approx 55.0\,V_y \text{ [N]}, \tag{17}$$

directed along $\pm$body-x. Saturation occurs when $\sqrt{2}F_{\perp,i} = N_i\mu$; with $N_{\min} = N_3 \approx 69.6$ N the lightest wheel saturates first:

$$V_{y,crit} = \frac{N_{\min}\mu(R-R_a)^2}{2p_2} \approx 0.63 \text{ m/s}, \tag{18}$$

$$V_{y,agg} = \frac{(\sum_i N_i)\mu(R-R_a)^2}{8p_2} \approx 0.79 \text{ m/s}. \tag{19}$$

Since $V_{y,crit} \propto \mu$, $V_{y,crit}(\mu) \approx 1.265\,\mu = \{0.51, 0.63, 0.76\}$ m/s for $\mu = \{0.4, 0.5, 0.6\}$ — the cleanest plant-vs-controller discriminator — and the Stribeck dip places the onset in the hysteretic band $[V_{y,crit}, sV_{y,crit}] \approx [0.63, 0.70]$ m/s.

3.3 Yaw-rate cap: the roller-sustaining drag $F_\perp$ fills the circle

A pure spin moves each contact tangentially. Its body-y component $\dot\psi\rho_{Xi} = \pm\dot\psi h$ is rolled out by the passive roller at $|\dot\gamma_i| = \sqrt{2}h\dot\psi/(R-R_a)$, costing

$$F_{\perp,i} = \frac{p_2\dot\gamma_i}{R-R_a} = \frac{\sqrt{2}\,p_2 h}{(R-R_a)^2}\dot\psi \approx 9.1\,\dot\psi \text{ N}, \tag{20}$$

the same passive force that set $V_{y,crit}$, now driven by the long spin lever h. It loads the circle directly. The roller-joint dissipation $d_\psi^{(p_2)} = 8p_2h^2/(R-R_a)^2 \approx 12.2$ is the power coefficient; the *contact* yaw moment of $F_\perp$ is only

$$\sum_i F_{\perp,i}\ell_{\perp,i} = \frac{4p_2h(h-l)}{(R-R_a)^2}\dot\psi \approx 2.2\,\dot\psi \text{ N m}, \tag{21}$$

at the short lever $\ell_\perp = (h-l)/\sqrt{2}$; the rest of $d_\psi^{(p_2)}$ is fed through the wheels and never sits on the circle. With the traction headroom

$$|F_{\parallel,i}| \le \sqrt{(\mu N_i)^2 - F_{\perp,i}^2}, \tag{22}$$

the band collapses as $F_\perp \to \mu N_i$, the lightest wheel pinching first:

$$\dot\psi_{\max} = \frac{\mu N_{\min}(R-R_a)^2}{\sqrt{2}\,p_2 h} = \frac{\sqrt{2}\,V_{y,crit}}{h} \approx 3.8 \text{ rad/s}. \tag{23}$$

The torque-only rate (≈ 19 rad/s) is never approached.

3.4 Acceleration headroom and the velocity–acceleration envelope

The caps of §3.1–3.3 are the $\dot{\vec{v}} = 0$ slice of a larger reachable set. To bound transient maneuvers we ask, at a velocity $\vec{v}$ already inside the envelope, how much acceleration $\dot{\vec{v}}$ the contacts can still supply on the *same* friction circle. The roller-frame split (7) answers it directly: $F_{\perp,i}(\vec{v})$ is velocity-slaved and consumes the circle radially, while $F_{\parallel,i}$ — the motor/traction component — must carry the entire inertial demand and lives in whatever headroom remains. Eq. (22) is therefore not just the yaw-cap pinch but a general per-wheel statement.

Per-wheel resolution. Write the bare-body balance (the contact form of (9), forces explicit) with the velocity-known drag moved to the right,

$$A_{\parallel}\,\vec{F}_{\parallel} = \underbrace{M\,\dot{\vec{v}} + C(\dot{\psi})\,\vec{v} - A_n\,\vec{F}_{\perp}(\vec{v})}_{\vec{b}(\vec{v},\dot{\vec{v}})}, \tag{24}$$

where the allocators carry each wheel's roller-frame force into the body wrench (F_x, F_y, M_z),

$$A_{\parallel} = \begin{bmatrix} \cos\delta_i \\ \sin\delta_i \\ \rho_{Xi}\sin\delta_i - \rho_{Yi}\cos\delta_i \end{bmatrix}_i^{\top}, \quad A_n = \begin{bmatrix} -\sin\delta_i \\ \cos\delta_i \\ \rho_{Xi}\cos\delta_i + \rho_{Yi}\sin\delta_i \end{bmatrix}_i^{\top}, \tag{25}$$

whose moment rows are the roller-frame arms $\ell_{\parallel,i} = \frac{h+l}{\sqrt{2}}\lambda_i$, $\ell_{\perp,i} = \frac{h-l}{\sqrt{2}}\nu_i$ of (39). Numerically,

$$A_{\parallel} = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 & 1 & 1 \\ -1 & 1 & 1 & -1 \\ -0.385 & 0.385 & -0.385 & 0.385 \end{bmatrix}, \quad A_{\parallel}^{+} = \begin{bmatrix} 0.354 & -0.354 & -0.918 \\ 0.354 & 0.354 & 0.918 \\ 0.354 & 0.354 & -0.918 \\ 0.354 & -0.354 & 0.918 \end{bmatrix}. \tag{26}$$

Min-norm allocation. $A_{\parallel}$ is 3×4 with $\ker A_{\parallel} = \frac{1}{2}(1,1,-1,-1)$ — the front/rear "squeeze" that produces zero body wrench but loads all four circles equally. The feedforward injects none of it, so we take the min-norm allocation $\vec{F}_{\parallel} = A_{\parallel}^{+}\vec{b}$, which gives the largest envelope; a factor of safety enters separately as $\mu N_i \to s\,\mu N_i$, $s \in [0.7, 0.8]$, shrinking the $F_{\perp}$ caps and $F_{\parallel}$ headroom together.

Bare vs. augmented mass — a circle-vs-torque caution. The contact-force balance (24) uses the *bare* body mass $M = \text{diag-block}(m_s, m_s, I_s)$ with the COM coupling, *not* the augmented M_{aug} of (9). The $4J_w/R^2$ and $4(l+h)^2 J_w/R^2$ reflections are wheel rotational inertia, spun up by motor torque through the wheel equation $J_w\ddot{\varphi}_i = M_i - p_1\dot{\varphi}_i - F_{x,i}R$; they load the actuator, not the contact. Using M_{aug} here would double-count the wheel inertia onto the circle and spuriously shrink the acceleration envelope by $\tilde{m}/m_s \approx 1.26$ on translation and $I_{\psi}/I_s \approx 1.31$ on yaw. This is the same p_1/p_2 ownership split as §2, now on the inertial side: $F_{\parallel}$ sees m_s; the motor sees $\tilde{m}$.

The envelope. The reachable set is the per-wheel headroom law, the acceleration generalization of (22),

$$\boxed{\left(A_{\parallel}^{+}\vec{b}\right)_i^2 + F_{\perp,i}(\vec{v})^2 \le (\mu N_i)^2, \qquad i = 1\ldots 4.} \tag{27}$$

The velocity-slaved drag fixes $F_\perp$ (radial); the inertial demand sets $F_\parallel$ (residual); the envelope shrinks to a point as $\vec{v}$ approaches the velocity caps. The drag-relocation vector $\Phi(\vec{v}) = A_n\vec{F}_\perp$ is velocity-only and linear,

$$\Phi(\vec{v}) = \left(0,\ \tfrac{2\sqrt{2}\,p_2}{(R-R_a)^2}V_y,\ \tfrac{2\sqrt{2}\,p_2 h(h-l)}{(R-R_a)^2}\dot{\psi}\right) = (0,\ 110.0\,V_y,\ 2.20\,\dot{\psi}), \tag{28}$$

its yaw entry exactly the contact yaw moment $2.2\,\dot{\psi}$ of (21) and its lateral entry the net strafe drag $2\sqrt{2}F_\perp$ of §3.2 — the velocity envelope re-entering as the affine offset of the acceleration problem.

Acceleration intercepts. At rest ($\vec{v} = 0$, $F_\perp = 0$, the full circle free) the pure-axis caps from (27) are

$$a_{x,cap} = 2.93, \quad a_{y,cap} = 2.86 \text{ m/s}^2, \quad \ddot{\psi}_{cap} = 9.62 \text{ rad/s}^2, \tag{29}$$

all pinched at the rear-left wheel ($N_{\min}$) and all below the aggregate isotropic value $\mu g = 4.9$ m/s^2, because a single wheel binds first under any feasible allocation. The longitudinal intercept is additionally torque-capped (§3.1).

Binding-wheel reduction (gating form). Across the operating box the rear-left wheel (#3, $\mu N_3 = 34.8$ N) is the binding constraint in every pure direction and in most mixed states, so the four-wheel test (27) collapses to a single scalar gate. With $\rho_{X,3} = -h$, $\rho_{Y,3} = +l$, and the third row of $A_\parallel^+ = (0.354, 0.354, -0.918)$,

$$F_{\perp,3} = \kappa\,(V_y - h\dot{\psi}), \qquad \kappa = \frac{\sqrt{2}\,p_2}{(R-R_a)^2} = 38.9, \tag{30}$$

$$F_{\parallel,3} = 0.354\,(b_x + b_y) - 0.918\,b_\Omega, \tag{31}$$

$$\text{feasible} \iff F_{\parallel,3}^2 \le (\mu N_3)^2 - F_{\perp,3}^2, \tag{32}$$

with $\vec{b}$ from (24) expanded as

$$b_x = m_s a_x - m a_Y \alpha - m_s\dot{\psi}V_y - m a_X\dot{\psi}^2, \tag{33}$$

$$b_y = m_s a_y + m a_X \alpha + m_s\dot{\psi}V_x - m a_Y\dot{\psi}^2 - 110.0\,V_y, \tag{34}$$

$$b_\Omega = -m a_Y a_x + m a_X a_y + I_s\alpha + m\dot{\psi}(a_X V_x + a_Y V_y) - 2.20\,\dot{\psi}, \tag{35}$$

($\dot{\vec{v}} = (a_x, a_y, \alpha)$). Only in the off-axis corners (large $|V_y|$ against opposing spin) does the front-left wheel (#1) bind first; the sampler should carry the #1 row as a second gate there, or fall back to the full (27). A coarse 729-node lookup of the maximum single-axis acceleration over the velocity grid accompanies these notes for rejection sampling.

3.5 Out-of-plane load transfer: the bound that sets the operating acceleration cap

The friction-circle caps of §3.4 ($a_{cap} \approx 2.9$ m/s^2) are derived in the planar 3-DOF model, where the per-wheel normal loads N_i are *static* (set once by weight and the COM offset, Table 1). A platform accelerating at a redistributes those loads through the out-of-plane pitch/roll the planar model omits. For a rigid (suspension-free) body the quasi-static per-wheel transfer is

$$\Delta N_i = \frac{m\,a\,z_{com}}{4\,d}, \qquad d = \begin{cases} h & \text{longitudinal} \\ l & \text{lateral,} \end{cases} \tag{36}$$

with z_{com} the COM height and $2h, 2l$ the wheelbase/track. Each contact force is *linear* in N_i ($\|\vec{F}_i\| \le \mu N_i$), so $\Delta N_i / N_i$ is exactly the fractional per-wheel force error the planar dataset cannot reproduce. The lateral axis binds ($l < h$) and the lightest wheel #3 ($N_3 = 69.6$ N, the friction-circle binding wheel of §3.3) is the worst case.

Holding this neglected effect below 5% bounds the operating acceleration. The plant carries no z-DOF, so z_{com} is a free parameter; for the youBot we take $z_{com} = R = 0.05$ m (COM at hub height), with $z_{com} = 2R$ as a sensitivity:

$$a_{5\%}^{lat} = \frac{0.05 \cdot 4l\, N_3}{m\, z_{com}} = \begin{cases} 1.39 \text{ m/s}^2 & z_{com} = R, \\ 0.70 \text{ m/s}^2 & z_{com} = 2R. \end{cases} \tag{37}$$

	$z_{com} = R$		$z_{com} = 2R$	
a [m/s^2]	$\Delta N / \bar{N}$	$\Delta N / N_3$	$\Delta N / \bar{N}$	$\Delta N / N_3$
0.5	1.4%	1.8%	2.9%	3.6%
1.0	2.9%	3.6%	5.7%	7.2%
1.5	4.3%	5.4%	8.6%	10.8%
2.0	5.7%	7.2%	11.5%	14.4%

Table 2: Per-wheel lateral load-transfer fraction (36) ($\bar{N} = 87.3$ N mean, $N_3 = 69.6$ N binding wheel).

We therefore cap the operating envelope at $a \le 1.0$ m/s^2, well inside the 2.9 m/s^2 friction-circle limit. At $z_{com} = R$ this holds the worst-wheel transfer to 3.6% (longitudinal 1.8%); at $z_{com} = 2R$ it reaches 7.2%, so $a \le 1.0$ is the appropriate, COM-height-aware cap for the youBot's low ($z_{com} \lesssim R$) centre of mass. This is the operational guarantee that the planar Julia dataset — and the high-fidelity engine benchmark validating it — never enters a regime where neglected load transfer exceeds the 5% fidelity budget.

4 Bounding envelope geometry ($\dot{\psi} = 0$)

The trackable-velocity set in the translational plane is neither a rhombus nor an ellipse. Each contact enforces an isotropic circle $\|\vec{F}_i\| \le N_i \mu$; since the slip components are affine in (V_x, V_y), each wheel's limit is an ellipse and the feasible set is the intersection of four oriented ellipses — a rounded convex region. The two axes are bounded by different mechanisms — a near-vertical torque-saturation wall in V_x, a rounded friction fold in V_y — breaking the 90° symmetry an ellipse would require. A compact description is an anisotropic superellipse

$$\left| \frac{V_x}{V_{x,cap}} \right|^n + \left| \frac{V_y}{V_{y,crit}} \right|^n = 1, \quad 1 < n < 2, \quad \frac{V_{x,cap}}{V_{y,crit}} \approx 7, \tag{38}$$

with the diamond-like skeleton ($n \to 1$) from the ±45° roller directions and the rounding ($n > 1$) from the LuGre smoothing. The envelope is three-dimensional: the coupled LuGre–Adamov law folds the spin $\omega_{z,i} = \dot{\psi}$ into the slip measure, so rising $\dot{\psi}$ eats the translational budget and the body is pinched in the combined-motion corners — the same pinch that §3.4 quantifies on the acceleration side.

5 Cross-axis coupling

5.1 Coupling channel

The off-centre COM makes M_{aug} non-diagonal, so a force residual on one axis leaks into the others through the off-diagonal entries of M_{aug}^{-1} (nonzero iff $a_X, a_Y \neq 0$). The per-wheel yaw contribution is $M_{z,i} = p_{x,i}F_{y,i} - p_{y,i}F_{x,i} + M_{z,i}^{spin}$.

5.2 Lateral axis (dominant)

In the roller frame $M_{z,i} = F_{\parallel,i}\ell_{\parallel,i} + F_{\perp,i}\ell_{\perp,i}$ with

$$\ell_{\parallel,i} = \frac{h+l}{\sqrt{2}}\lambda_i \ (\lambda = (-,+,-,+)), \qquad \ell_{\perp,i} = \frac{h-l}{\sqrt{2}}\nu_i \ (\nu = (+,+,-,-)), \tag{39}$$

so the traction moment acts at the long arm 0.272 m and the roller-drag moment at the short arm 0.060 m — a 4.5× leverage advantage for $F_\parallel$. The moment is exactly zero up to and including $V_{y,crit}$ (load-independent symmetric forces) and ramps from zero only above it,

$$M_z^{lat} = \mu\, mg\, a_X \frac{l}{h} \approx 1.50 \text{ N m} \ \longrightarrow\ \mu\, mg\, a_X \frac{h+l}{\sqrt{2}\,h} \approx 2.7 \text{ N m}, \tag{40}$$

as the roller-spin cap clips the weak-lever cancelling term. This routes the front–rear a_X imbalance through $F_\parallel$, which — via M_{aug}^{-1} — contaminates V_x.

5.3 Longitudinal axis (present but subdominant)

Longitudinal drive needs no roller spin ($F_{\perp,i} = 0$), so each saturated force lies along its axle at the full traction arm with no cancelling term:

$$M_z^{lon} = \mu\, mg\, a_Y \frac{h+l}{\sqrt{2}\,l} \approx 6.94 \text{ N m}. \tag{41}$$

Its bare magnitude is the largest of the three, yet the axis is benign operationally: V_x friction-saturates only transiently, the circle is otherwise empty, and the kick is self-limiting.

5.4 Yaw-rate axis: centrifugal creep coupling (benign)

The centrifugal terms in (10) are a body force $\vec{F}_{cf} = m\dot{\psi}^2(a_X, a_Y)$, $\|\vec{F}_{cf}\| = m|a|\dot{\psi}^2$, $|a| = 0.031$ m, held by the translational friction until $\dot{\psi}_{cf} = \sqrt{\mu N_{tot}/(m|a|)} \approx 14$ rad/s. Because $\dot{\psi}_{cf} \gg \dot{\psi}_{\max}$ the platform cannot reach centrifugal gross slip. Below the cap the coupling is a small steady creep $v_{creep} \approx m|a|\dot{\psi}^2/d_t \approx 1.2$ cm/s at $\dot{\psi} \approx 2$. Yaw remains the most benign axis.

5.5 Role of the COM offset: imperfect pitchfork

Every coupling moment is $\propto$ the COM offset and vanishes for $a_X = a_Y = 0$; simultaneously M_{aug} becomes diagonal and the leakage channel closes. With a centred COM the strafe-saturation is symmetric and the relevant bifurcation is a pitchfork; the COM offset is the imperfection parameter that unfolds it into a saddle-node, preselecting the yaw direction.

6 Post-cap behaviour: yaw asymmetry and cross-axis leak

6.1 Directional (CW/CCW) asymmetry of the yaw cap

Every friction drag is even in $\dot{\psi}$; the COM offset supplies the only odd-in-$\dot{\psi}$ coupling, via (i) a fixed centripetal $-m(a_X, a_Y)\dot{\psi}^2$ added to a sign-flipping tangential demand $\pm\mu N_i \hat{t}_i$, which add at one spin sense and cancel at the other, and (ii) the gyroscopic yaw term $m\dot{\psi}|a|(\hat{a}\cdot\vec{V})$. The asymmetry scales as $m|a|\dot{\psi}^2/(4\mu N_{\min}) \approx 10\%$ at $\dot{\psi} \approx 4$ and vanishes at $a_X = a_Y = 0$.

6.2 Why exceeding $V_{y,crit}$ bleeds into $\dot{\psi}$ and V_x

$V_{x,cap}$ is soft (torque/drag): past it the circle stays empty and the platform merely lags, on-axis and quiet. $V_{y,crit}$ is a hard friction-circle wall: past it the passive roller can no longer sustain $F_\perp$, $V_{py,i}$ grows, and the saturated force caps and rotates. The load spread breaks symmetry, the yaw moment climbs from 0 at the cap toward 1.5–2.7 N m, forcing $\dot{\psi} \neq 0$; the induced $\dot{\psi}$ re-enters V_x through the Coriolis $-m_s\dot{\psi}V_y$ and the $-ma_Y$ off-diagonal. With d_y lost,

$$\tilde{m}\dot{V}_x = -d_xV_x + m_s\dot{\psi}V_y, \qquad \tilde{m}\dot{V}_y = -\underbrace{d_yV_y}_{\to 0} - m_s\dot{\psi}V_x, \tag{42}$$

so the skew $m_s\dot{\psi}$ whirls V_y into V_x and back, a bounded-then-diverging lateral/yaw oscillation. A longitudinal over-command does none of this — the asymmetry is the soft-vs-hard cap distinction.

6.3 Exceeding $\dot{\psi}_{\max}$: leak into V_x and V_y

The dual of §6.2. Past $\dot{\psi}_{\max}$ the lightest wheel cannot supply $F_\perp$ and slides along $\hat{n}$, delivering only $\mu N_{\min}$; the symmetric balance breaks and a net body force appears,

$$\vec{F}_{leak} \approx \Delta F\left(\tfrac{1}{\sqrt{2}}, -\tfrac{1}{\sqrt{2}}\right), \quad \Delta F = F_\perp^{dem} - \mu N_{\min}, \tag{43}$$

(forward-right for $\dot{\psi} > 0$; flips for $\dot{\psi} < 0$). Since $F_\parallel \to 0$ at the saturated wheels, the traction that drove the spin vanishes ($\dot{\psi}$ plateaus) and the drift is uncorrectable at the very wheels producing it — the likely origin of the observer's actuator saturation at heading-ramp onset, motivating a per-axis split $\omega_\psi \gg \omega_{xy}$.

7 Summary of closed-form results

A Equation appendix — Word-compatible LaTeX commands

Each block is the linear LaTeX string for the corresponding numbered equation, restricted to commands accepted by the Word equation editor. Concatenate split lines before pasting. Entries (E49)–(E55) are the velocity–acceleration envelope of §3.4.

```
(E13) Lateral velocity cap (onset, lightest wheel):
V_{y,crit} = \frac{ N_{min}\, \mu\, (R - R_a)^2 }{ 2 p_2 }

(E33) Yaw-rate cap (roller-sustaining drag F_perp fills the circle):
F_{\perp,i} = \frac{ \sqrt{2}\, p_2 h }{ (R - R_a)^2 }\, \dot{\psi}
\dot{\psi}_{max} = \frac{ \mu N_{min} (R - R_a)^2 }{ \sqrt{2}\, p_2 h } = \frac{ \sqrt{2}\, V_{y,
crit} }{ h }
```

Quantity	Expression	Value (case 1, $\mu = 0.5$)
Longitudinal speed cap	$\tau_{\max} R/p_1$	4.55 m/s
Longitudinal accel. cap (aggregate)	$(\sum N_i)\mu/m_s$	4.9 m/s^2
Lateral onset (1st wheel)	$N_{\min}\mu(R-R_a)^2/(2p_2)$	0.63 m/s
Lateral full saturation	$(\sum N_i)\mu(R-R_a)^2/(8p_2)$	0.79 m/s
Yaw-rate cap ($F_\perp$ fills circle)	$N_{\min}\mu(R-R_a)^2/(\sqrt{2}p_2 h)$	$\approx$ 3.8 rad/s
Velocity–acceleration envelope (§3.4, per-wheel, bare mass, RL-bound)		
Long. accel. cap (per-wheel)	(μN_3) free at $\vec{v}=0$	2.93 m/s^2
Lateral accel. cap (per-wheel)	(μN_3) free at $\vec{v}=0$	2.86 m/s^2
Yaw accel. cap (per-wheel)	(μN_3) free at $\vec{v}=0$	9.62 rad/s^2
Drag-relocation $\Phi(\vec{v})$	$(0,\, 2\sqrt{2}p_2 V_y/(R-R_a)^2,\, 2.2\dot{\psi})$	$(0, 110V_y, 2.2\dot{\psi})$
Master envelope	$(A_\parallel^{+}\vec{b})_i^2 + F_{\perp,i}^2 \le (\mu N_i)^2$	—
Out-of-plane load-transfer bound (§3.5)		
Per-wheel load transfer	$m\, a\, z_{com}/(4d)$	$d=h$ long., l lat.
Operating accel. cap (<5% transfer)	$0.05 \cdot 4lN_3/(mz_{com})$	1.0 m/s^2 ($z_{com}\!=\!R$)
Contact yaw moment of $F_\perp$	$4p_2h(h-l)/(R-R_a)^2$	$\approx$ 2.2 N m s
CW/CCW cap asymmetry	$m\lvert a\rvert\dot{\psi}^2/(4\mu N_{\min})$	$\approx$ 10%
Yaw centrifugal threshold	$\sqrt{\mu N_{tot}/(m\lvert a\rvert)}$	14 rad/s
Anisotropy ratio	$V_{x,cap}/V_{y,crit}$	$\approx$ 7
Sustained yaw moment (lat.)	ramp $0 \to \mu mga_X(h+l)/(\sqrt{2}h)$	0 – 1.5–2.7 N m
Transient yaw moment (lon.)	$\mu mga_Y(h+l)/(\sqrt{2}l)$	6.94 N m

```
| F_{\parallel,i} | \le \sqrt{ ( \mu N_i )^2 - F_{\perp,i}^2 } (traction headroom)

(E45) Roller-frame force decomposition (axle e_i, rolling dir n_i; friction circle):
\vec{F}_i = F_{\parallel,i}\, \hat{e}_i + F_{\perp,i}\, \hat{n}_i,\quad
\hat{e}_i = (\cos\delta_i, \sin\delta_i),\ \hat{n}_i = (-\sin\delta_i, \cos\delta_i)
F_{\parallel,i}^2 + F_{\perp,i}^2 \le ( \mu N_i )^2
F_{\perp,i} = p_2\, \dot{\gamma}_i / (R - R_a)

(E46) Roller-frame yaw arms (traction lever vs roller-drag lever):
\ell_{\parallel,i} = \frac{h+l}{\sqrt{2}}\, \lambda_i,\quad \lambda = (-,+,-,+)
\ell_{\perp,i} = \frac{h-l}{\sqrt{2}}\, \nu_i,\quad \nu = (+,+,-,-)

================ velocity-acceleration envelope (Sec 3.4) ================

(E49) Per-wheel acceleration headroom (master velocity-acceleration envelope):
( A_\parallel^{+}\, \vec{b} )_i^2 + F_{\perp,i}(\vec{v})^2 \le ( \mu N_i )^2,\quad i = 1..4
\vec{b}(\vec{v},\dot{\vec{v}}) = M\, \dot{\vec{v}} + C(\dot{\psi})\, \vec{v} - A_n\, \vec{F}_\
perp(\vec{v})

(E50) Free-component (traction) allocator and its min-norm pseudo-inverse:
A_\parallel = \begin{bmatrix} \cos\delta_i \\ \sin\delta_i \\ \rho_{Xi}\sin\delta_i - \rho_{Yi}\
cos\delta_i \end{bmatrix}_i^{\top}
\ker A_\parallel = \tfrac{1}{2}(1,1,-1,-1),\qquad \vec{F}_\parallel = A_\parallel^{+}\, \vec{b}

(E51) Drag-relocation operator and closed-form offset (velocity-only, linear):
A_n = \begin{bmatrix} -\sin\delta_i \\ \cos\delta_i \\ \rho_{Xi}\cos\delta_i + \rho_{Yi}\sin\
delta_i \end{bmatrix}_i^{\top}
\Phi(\vec{v}) = A_n \vec{F}_\perp = \left( 0,\ \frac{2\sqrt{2}\, p_2}{(R-R_a)^2} V_y,\ \frac{2\
sqrt{2}\, p_2 h (h-l)}{(R-R_a)^2} \dot{\psi} \right)

(E52) Bare-vs-augmented mass caution (circle uses bare M; motor sees M_aug):
```

```
A_\parallel \vec{F}_\parallel = M\, \dot{\vec{v}} + \ldots,\qquad M = \mathrm{diag\text{-}block}(
m_s, m_s, I_s)\ \text{with COM coupling}
\text{wheel inertia } 4 J_w / R^2 \to \text{motor torque (E4), NOT contact circle}

(E53) Acceleration intercepts from rest (v = 0, F_perp = 0, full circle free):
a_{x,cap} = 2.93,\quad a_{y,cap} = 2.86\ \mathrm{m/s^2},\quad \ddot{\psi}_{cap} = 9.62\ \mathrm{
rad/s^2}
\text{all pinched at } N_{min}\ (\text{rear-left}),\quad < \mu g = 4.9\ \mathrm{m/s^2}\ (\text{
aggregate})

(E54) Rear-left (#3) binding-wheel scalar gate:
\kappa = \frac{\sqrt{2}\, p_2}{(R - R_a)^2} = 38.9
F_{\perp,3} = \kappa ( V_y - h\, \dot{\psi} )
F_{\parallel,3} = 0.354 ( b_x + b_y ) - 0.918\, b_\Omega
\text{feasible} \iff F_{\parallel,3}^2 \le ( \mu N_3 )^2 - F_{\perp,3}^2,\quad \mu N_3 = 34.8

(E55) Demand vector components b(v, vdot) for the gate (vdot = (a_x, a_y, alpha)):
b_x = m_s a_x - m a_Y \alpha - m_s \dot{\psi} V_y - m a_X \dot{\psi}^2
b_y = m_s a_y + m a_X \alpha + m_s \dot{\psi} V_x - m a_Y \dot{\psi}^2 - 110.0\, V_y
b_\Omega = -m a_Y a_x + m a_X a_y + I_s \alpha + m \dot{\psi} ( a_X V_x + a_Y V_y ) - 2.20\, \dot
{\psi}

(E56) Factor of safety (applied to the whole circle, both components):
\mu N_i \to s\, \mu N_i,\quad s \in [0.7, 0.8]

(E57) Out-of-plane per-wheel load transfer (sets the a <= 1.0 m/s^2 operating cap):
\Delta N_i = \frac{ m\, a\, z_{com} }{ 4 d },\quad d = h\ (long.)\ |\ l\ (lat.)
a_{5\%}^{lat} = \frac{ 0.05 \cdot 4 l N_3 }{ m\, z_{com} } = 1.39\ (z=R)\ |\ 0.70\ (z=2R)\ \
mathrm{m/s^2}
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Reference. Adamov, B. I., & Saipulaev, G. R. (2020). Research on the dynamics of an omnidirectional platform taking into account real design of mecanum wheels (as exemplified by KUKA youBot). *Russian Journal of Nonlinear Dynamics*, 16(2), 291–307.